



Fig. 6: A digital polygraph employs transducers to convert the energy of the displaced air into electronic signals

attached to two of the subject's fingers. These plates measure the skin's ability to conduct electricity. When the skin is hydrated (as with sweat), it conducts electricity much more easily than when it is dry.

Arm and leg movements. Some polygraphs also record arm and leg movements. As the examiner asks ques-

increasing the psychological stress of a subject under interrogation, but galvanic skin response and heart rate have little to do with the process of lying. Thus, it is acknowledged that a habitual liar can fool a polygraph machine and examiner.

But, lie detectors equipped with nuclear magnetic-resonance brain

tions, signals from the sensors connected to the body are recorded on a single strip of moving paper.

Are the results really accurate?

Some experts say that lie detectors might have some vague use in

scans are expected to become more accurate devices. Furthermore, forget the elevated pulses, sweaty palms and respiration changes scrutinised by conventional lie detector tests. There is a more direct and, in theory at least, accurate way to measure deceit: track the flow of blood where the lie is born. Lying requires an extra bit of thought, which pulls more blood into a swath of the brain just beneath the forehead. These flows can be tracked optically.

Biophysicist Britton Chance—who 60 years ago was part of the wartime research team that developed radar at MIT's Radiation Laboratory—is pioneering a technology that can literally see a lie as it is spoken. He believes this method is better than conventional tests because the flows can't be suppressed and are less likely to have been caused by the stress of test taking. ●

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